

Chronic Inflammatory Demyelinating Polyradiculoneuropathy (CIDP): A Review of Clinical Syndromes and Treatment Approaches in Clinical Practice

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Abstract

Chronic inflammatory demyelinating polyneuropathy (CIDP) is a chronic, acquired immune and inflammatory disorder of the peripheral nervous system. The classic form of the disorder is manifested by progressive or relapsing proximal or generalized limb weakness and areflexia, and usually easily recognized; it is the large number of regional and functional variants and variety of associated illnesses that pose a challenge to the clinician in practice. Similarly, laboratory and electromyography criteria have been developed to confirm the diagnosis; however, these various schemes are contrived because only 50% to 60% of patients with typical clinical features of CIDP fulfill these strict electrodiagnostic research criteria. Several studies have established the efficacy of immune therapies such as corticosteroids, plasma exchange, and intravenous immune globulin as the mainstay of treatment of CIDP, but these treatments might provide only short-term benefit. This review offers an approach to the evaluation and management of patients with CIDP and highlights the difficult clinical problems in those who do not respond or frequently relapse after treatment with standard therapies such as patients with CIDP and concomitant axonal loss, and the assessment of those with CIDP and concurrent diseases such as diabetes mellitus.

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Chronic inflammatory demyelinating polyneuropathy (CIDP) is a chronic, acquired immune and inflammatory disorder of the peripheral nervous system.^{1–9} The pathogenesis of CIDP is incompletely understood and might encompass several humoral and cell-mediated mechanisms. Immunopathologic studies have shown that, in at least the common variety, there is endoneurial inflammation and an immune-mediated attack directed against components of the myelin sheath.^{8–13} The classic form of the disorder is characterized by: 1) progressive limb weakness, sensory loss and areflexia with a relapsing or progressive course; 2) electrophysiological abnormalities of demyelination, including prolonged distal and F-wave latencies, reduced conduction velocity, and conduction block/temporal dispersion; 3) laboratory features of albumino-cytologic dissociation in cerebrospinal fluid (CSF); and 4) inflammation and demyelination and remyelination on nerve biopsy.^{1–9} Numerous small empiric trials have established the short-term efficacy

of immune therapies such as corticosteroids, plasma exchange, and intravenous immune globulin (IVIG) as the mainstay of treatment of CIDP,^{14–22} with several reports indicating the promise of novel therapies.^{23,24} Although guidelines for the diagnosis and treatment of CIDP have been developed,^{1,2,7–9} the large variety of clinical and electrophysiological variants, associated systemic conditions, and lack of sustained improvement with standard treatments in up to 30% of patients provide challenges to the clinician in practice. This review offers the current conventional views of the typical and variant clinical presentations, the laboratory and electrodiagnostic features, pathology, prognosis, and treatment of CIDP gleaned mainly from controlled, randomized trials and experience with large numbers of personally treated patients.

Clinical Features

CIDP is one in a spectrum of acquired immune-mediated demyelinating neuropathies that differ from each other, mainly in their time course. The 2 traditional categories of acute inflammatory demyelinating neuropathy (AIDP, Guillain-Barré syndrome) and CIDP have similar presentations except that the symptoms peak in less than several weeks in AIDP and progress for longer than 2 months in

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CIDP. For an illness that reaches its nadir between 4 weeks and 2 months, the term subacute demyelinating neuropathy has been used.²⁵ The natural history of the latter group can follow the pattern of either AIDP (monophasic course with a plateau and then recovery) or CIDP (progressive or relapsing). In large series, CIDP represents approximately 20% of initially undiagnosed neuropathies and accounts for approximately 10% of all patients referred to neuromuscular clinics.² The disorder most commonly occurs in adults between 40 and 60 years, but can affect the elderly and children.^{2–6,26–28} There is a slight predilection for men. The prevalence of CIDP is approximately 1.2 cases per 100,000 individuals, comparable to the frequency of AIDP.²⁹ Two thirds of cases are progressive and the remainder have a relapsing course, although recent experience suggests that relapses are often closely linked to withdrawal of therapy.^{3–6} The relapsing form tends to affect younger individuals.²⁶

The initial symptoms of CIDP are progressive, symmetric limb weakness and sensory loss that usually begins in the legs. Patients report difficulty walking, climbing stairs, and arising from a chair, and might have falls, or they might report reduced manual dexterity (trouble buttoning shirts, pulling up zippers, and so on) if hand weakness is present. The majority of patients also have sensory features, with many reporting numbness, tingling, or buzzing of the acral extremities (paresthesias) and gait unsteadiness reflecting sensory ataxia. Neuropathic pain occurs in a few.³ The examination shows symmetric proximal and distal limb weakness, generalized hypo- or areflexia, and distal sensory loss with large-fiber modalities (joint position, vibration) being disproportionately affected.^{1–6,26,27,30}

A useful clinical sign of CIDP is the discrepancy between the degree of weakness and lack of atrophy in an affected muscle group; this strongly implicates nerve demyelination rather than axonal loss. Facial, oropharyngeal, and ocular involvement occur in less than 15% of patients, and autonomic dysfunction and ventilatory failure develop in fewer than 10% of cases.^{3–5,26,27} Unusual features associated with CIDP, mostly in single case reports, include prominent vertigo,³¹ papilledema,^{4,26} and head drop.³²

Clinical Variants

Several reports have highlighted a variety of unusual clinical patterns that probably represent variants of CIDP and are accepted under the same umbrella term because they share most of the clinical, electrophysiological, and spinal fluid findings (Table 1). For example, like AIDP, certain forms can be classified as “functional” variants, with pure motor, pure sensory, and ataxic patterns.^{3,33–37} In several large series, up to 15% of patients have a pure sensory syndrome^{3,37}; others have a pure motor demyelinating neuropathy that can be distinguished from multifocal motor neuropathy by the presence of generalized proximal and

Table 1. Clinical variants of CIDP

Temporal course
Subacute inflammatory demyelinating polyneuropathy (SIDP)
Relapsing Guillain-Barré syndrome pattern
Severity
“Minimal” CIDP
Regional variants
Distal pattern (DADS)
Paraparetic CIDP
Focal or multifocal CIDP (MADSAM)
Isolated cranial neuropathies
Functional variants
Pure motor CIDP
Pure sensory CIDP
Ataxic CIDP

Abbreviations: CIDP, chronic inflammatory demyelinating polyneuropathy; DADS, distal acquired demyelinating symmetric neuropathy; MADSAM, multifocal acquired demyelinating sensory and motor neuropathy.

distal weakness, relative symmetry, and response to corticosteroids.^{3,38} In contrast, a number of “regional” variants also have been recognized, including a multifocal form that simulates mononeuritis multiplex,^{39–41} an upper limb demyelinating neuropathy that might be strikingly focal,^{41,42} and a paraparetic form associated with nerve root hypertrophy and symptoms suggestive of lumbar stenosis.^{43,44} A few patients have prominent or isolated cranial nerve palsies.^{4,37,45,46} Katz and colleagues stressed a rare, distal, symmetric, predominantly sensory demyelinating polyneuropathy that was quite similar to the polyneuropathy associated with an IgM monoclonal protein and elevated antimyelin-associated glycoprotein antibodies.⁴⁷

Other similar illnesses have been distinguished from typical CIDP based on the temporal course or severity of the disorder. For example, Hughes and colleagues and others have identified patients with a time course between AIDP and CIDP, as noted previously.^{25,48} Occasionally, the initial manifestations of CIDP might mimic the acute onset of Guillain-Barré syndrome followed by slower progression, or with a chronic, relapsing course;^{3,49–51} Grand’Maison and coworkers described 12 patients with a chronic, recurrent form of Guillain-Barré syndrome that likely represents a type of CIDP.⁵¹ Uncini and associates described 3 patients with mild distal paresthesias, demyelinating nerve conduction studies, and elevated cerebrospinal fluid protein level, and suggested the term “minimal CIDP” to reflect the limited nature of the illness.⁵² These regional and functional variants seem to respond to the conventional therapies for CIDP with about the same frequency (see subsequently in this article), but have never been subjected to carefully controlled study.

Chronic Inflammatory Demyelinating Polyradiculoneuropathy and Concurrent Diseases

A number of medical conditions have occurred simultaneously with CIDP and are implicated in its pathogenesis. For example, most large series of CIDP have had approximately 20% to 30% of patients with a monoclonal gammopathy detected by serum immunofixation.^{1-3,30,53} Some of these patients are found to have a malignant plasma cell dyscrasia (multiple myeloma, POEMS syndrome [Polyneuropathy, Organomegaly, Endocrinopathy, M-spike, Skin changes], lymphoma, or Waldenström's macroglobulinemia), but the majority have a monoclonal gammopathy of undetermined significance (MGUS).^{3,54-56} These patients generally are managed no differently than those with classic CIDP if a plasma cell neoplasm has been excluded. Most patients with the CIDP-MGUS combination and IgG or IgA monoclonal proteins have a disorder that is indistinguishable from idiopathic CIDP, including improvement with conventional therapies.^{1-3,53-58} In contrast, those with IgM-MGUS usually have greater sensory loss, tremor, more severe demyelinating abnormalities on electromyography (EMG) studies, an increased frequency of antimyelin-associated glycoprotein (anti-MAG) or other pathogenic autoantibodies, and are apparently less responsive to conventional immunotherapies for CIDP.^{54-56,59,60} This latter group probably constitutes a different disease.

Approximately 10% to 20% of patients with CIDP have an associated systemic medical disorder, although the precise relationship of the underlying illness to the neuropathy varies from case to case (Table 2). In essence, these are "secondary symptomatic" inflammatory demyelinating neuropathies in distinction to the idiopathic variety. These conditions include several autoimmune or connective tissue disorders (systemic lupus erythematosus, thyroid disease, rheumatoid arthritis, Sjögren's syndrome, myasthenia gravis, and sarcoidosis), chronic infections (HIV, HTLV I, hepatitis), organ transplantation, various malignancies, diabetes, renal insufficiency, medications (procainamide, interferon alpha), vaccination, CIDP superimposed on a hereditary neuropathy, and CIDP associated with demyelinating disorders of the central nervous system.^{7,8, 61-73} It is not known if one of these concurrent illnesses alters the response to therapy or the prognosis of patients with CIDP. However, identifying these associated conditions could facilitate the initial choice of therapy (eg, corticosteroids in patients with CIDP and systemic lupus erythematosus) and has important implications for the patient (eg, CIDP associated with HIV infection or lymphoma). In many cases of demyelinating polyneuropathy with systemic illness, the pathobiology of nerve damage differs from CIDP, but the clinical and electrophysiological features overlap.

Chronic Inflammatory Demyelinating Polyradiculoneuropathy and Diabetes Mellitus

Patients with diabetes mellitus and a polyneuropathy frequently have elements of demyelination on EMG studies, and it is well known that many have an elevated cerebrospinal fluid protein level. Nerve biopsy might reveal modest demyelinating changes and even inflammatory cell infiltration. Diabetic lumbosacral radiculoplexopathy might share pathologic features with CIDP. Our approach has been to attribute the illness to CIDP if the clinical and laboratory features (EMG and spinal fluid studies) are supportive, irrespective of a diagnosis of concurrent diabetes mellitus.^{63,64} We base this on the fact that most patients with a distal diabetic polyneuropathy do not have prominent proximal or generalized weakness and do not have a progressive course over a few months, and that conduction block (at sites not prone to entrapment) on EMG studies is rare in diabetic neuropathy, and, in our experience, a spinal fluid protein level greater than 100 mg/dL is unusual in patients with diabetic neuropathy. Nevertheless, a therapeutic trial of IVIG or plasma exchange could be used in some cases, and in our view a substantial and rapid clinical improvement supports the diagnosis of CIDP.⁶⁴

Laboratory Evaluation

The laboratory tests that assist in evaluating the patient with CIDP include complete blood count, serum chemistries, blood urea nitrogen, creatinine, HIV titer, fasting blood glucose and glycosylated hemoglobin, antinuclear antibodies, and, perhaps most importantly, serum and urine immunofixation electrophoresis for detection of a paraprotein.^{1,2} The detection of a monoclonal protein warrants investigation for a malignant plasma cell dyscrasia before concluding the M-spike is of lesser significance; an M-protein greater than 3.0 g/dL is almost always indicative of a malignant plasma cell disorder, whereas a level less than 1.5 g/dL is more often observed in patients who have MGUS. When an M-protein has been detected, the serum also should be evaluated for the presence of cryoglobulins and β -2 microglobulin; serum viscosity should be measured when >3.0 g/dL of IgM monoclonal protein is detected. A bone scan and bone marrow biopsy are necessary in patients who have >3.0 g/L of monoclonal protein and systemic features suggesting multiple myeloma (fatigue, bone pain, weight loss, anemia, hypercalcemia, renal failure). A skeletal bone survey also is required in most cases to exclude bone lesions associated with osteosclerotic myeloma and the POEMS syndrome, especially if there is an IgG, IgA, or lambda light chain monoclonal protein. Suspicious sclerotic bone lesions should be biopsied. Chest and abdominal computed tomography scanning is useful in patients suspected of having lymphoma or the POEMS syndrome associated with Castleman's disease.

Table 2. Concurrent diseases associated with inflammatory demyelinating polyneuropathy (“secondary” CIDP)

Chronic infections	Hereditary neuropathy (CIDP and CMT)
HIV infection	
HTLV I	
Lyme disease	Medications
Cat scratch disease	Interferon alpha
Epstein-Barr infection	Procainamide
Hepatitis C	Tacrolimus
Connective tissue and autoimmune disorders	Vaccinations
Systemic lupus erythematosus	
Sjogren syndrome	Solid organ transplantation
Rheumatoid disease	
Giant cell arteritis	
Inflammatory bowel disease	
Myasthenia gravis	
Chronic active hepatitis	
Multiple sclerosis	
Systemic medical disorders	
Diabetes mellitus	
Thyrotoxicosis	
Chronic renal failure requiring dialysis	
Membranous glomerulonephropathy	
Malignancy	
Hepatocellular carcinoma	
Melanoma	
Pancreatic carcinoma	
Colon adenocarcinoma	
Lymphoma	
Paraneoplastic demyelinating neuropathy	
Paraprotein associated disorders	
Monoclonal gammopathy of undetermined significance (MGUS)	
Osteosclerotic myeloma (POEMS syndrome)	
Multiple myeloma	
Waldenstrom macroglobulinemia	
Amyloidosis	
Castleman’s disease	

Abbreviations: CMT, Charcot-Marie-Tooth disease; MGUS, monoclonal gammopathy of undetermined significance; POEMS, Polyneuropathy, Organomegaly, Endocrinopathy, M-spike, Skin changes.

Additional laboratory studies that might be indicated in certain cases with features suggestive of a connective tissue disorder include serum ANCA level, rheumatoid factor, angiotensin-converting enzyme, anti-SSA/SSB antibodies, hepatitis B and C, HTLV I, and Lyme serologies. We usually obtain genetic studies for Charcot-Marie-Tooth disease type IA, IB, and X in patients suspected of having an inherited neuropathy and in young patients without unequivocal conduction block on electrodiagnostic studies. Brain and spinal cord magnetic resonance imaging studies also should be obtained when the neurologic history or

findings suggest concomitant central nervous system demyelination. In those individuals with constitutional features (unexplained weight loss, fevers, adenopathy, anemia, and so on), a search for a malignancy is warranted.

A lumbar puncture also is indicated in patients suspected of having CIDP; an elevated CSF protein level (>45 mg/dL) is detected in at least 80% of patients.^{1-6,8,26,27} However, the diagnosis of CIDP should not be excluded based only on a normal CSF protein concentration. The cell count is usually normal, but as many as 10% of patients have greater than 5 lymphocytes/mm³. The Ameri-

can Academy of Neurology (AAN) criteria have indicated that there should be fewer than 10 cells in the spinal fluid and fewer than 50 cells in patients with HIV infection.⁷ Accordingly, the presence of a CSF pleocytosis should prompt an evaluation for HIV infection, as well as Lyme disease, lymphomatous meningitis, and sarcoidosis. Oligoclonal bands are found in the CSF in approximately 65% of cases and do not appear to be of any consequence.⁷⁴

Electrodiagnostic Criteria

Nerve conduction studies establish the diagnosis of CIDP with certainty in the majority of cases. Several electrodiagnostic criteria have been proposed, all requiring some combination of: 1) reduced conduction velocities (eg, <80% of the lower limit of normal [LLN] if the distal motor amplitude is normal, and <70% of LLN if the amplitude is substantially reduced); 2) prolonged distal motor latencies, 3) prolonged F-wave latencies (eg, >125% of the upper limit of normal [ULN] if the distal motor amplitude is normal, and >150% of ULN if the amplitude is reduced for distal latencies and F-waves); and 4) conduction block/temporal dispersion (eg, conduction block is >50% reduction of proximal/distal [p/d] amplitude and abnormal temporal dispersion is >130% increase of p/d duration).^{7,22,26,30,37,75-78} According to the AAN criteria, the presence of 3 of the 4 criteria in 2 or more nerves establishes the electrophysiological diagnosis of CIDP for research purposes.⁷ For the practicing clinician, these various schemes are contrived because only 50% to 60% of patients with typical clinical features of CIDP fulfill these strict electrodiagnostic criteria (see subsequently in this article).⁷⁹ Accordingly, numerous alternative EMG criteria have been offered that are more sensitive without sacrificing specificity.^{22,30,77,78} A thoughtful comparison of the advantages and disadvantages of each of the proposed criteria has been given recently by Sander and Latov.⁸⁰

Because demyelination in CIDP is frequently a multifocal process, EMG sampling of multiple nerves (at least 4, and probably more) is necessary. Similarly, proximal nerve and root stimulation might be required in some cases to confirm proximal conduction block when this finding is not detected in distal segments.⁸¹ Care must be taken to ensure that nerve stimulation is supramaximal. Computer simulation studies have demonstrated that an amplitude reduction as great as 50% between proximal and distal points of stimulation might occur in normal motor nerves, a phenomenon termed "pseudo-conduction block." This has been attributed to interphase cancellation of motor potential waveforms.⁸² Therefore, the criterion for definite conduction block has been increased from 20% (initial AAN criteria) to a 50% amplitude reduction following proximal nerve stimulation.⁷⁶

Abnormalities of the sensory nerve action potentials (SNAPs) have not been addressed in most of the published EMG criteria for CIDP. Many patients have low or absent SNAPs. When sensory potentials are obtainable, the distal

latencies or conduction velocities are usually slowed and in the demyelinating range.^{5,26,79} A characteristic finding of absent or reduced median or ulnar SNAPs with normal sural sensory potentials ("sural sparing") indicates a non-length-dependent pattern and is suggestive of demyelinating neuropathy.

Needle electrode examination in patients with a pure demyelinating form of CIDP shows reduced recruitment patterns of motor units without active or chronic denervation. The presence of abnormal spontaneous activity (fibrillation potentials) or large-amplitude, long-duration motor unit potentials indicating chronic partial denervation and reinnervation signifies an axonal component; these axonal changes are present in varying degrees in approximately two thirds of cases.^{1-3,26}

Electrodiagnostic studies have several important shortcomings in establishing the diagnosis of CIDP: 1) patients might accrue substantial axonal loss with absent or greatly reduced motor potentials, and therefore nerve conduction abnormalities might not fulfill criteria for demyelination; 2) those with a pure sensory variant might have absent SNAPs and relatively normal motor potentials, but most have demyelinating features on motor studies despite the absence of weakness; 3) a few patients have electrodiagnostic abnormalities limited to proximal motor nerve conduction block that are detected only with motor root or Erb's point stimulation⁸¹; and 4) in clinical practice, there are a large group of patients who do not fulfill EMG research criteria for CIDP despite extensive nerve sampling, as previously noted.^{79,80} In these patients, spinal fluid analysis and nerve biopsy might be especially helpful to confirm the diagnosis, and a therapeutic trial might be warranted in patients who do not fulfill electrodiagnostic criteria.^{1-3,22,26,30,80}

Pathology

The classic pathologic features of CIDP include demyelination, remyelination (onion bulbs), endoneurial edema, and inflammatory cell infiltrates in the epineurium and endoneurium, usually with preferential involvement of the nerve roots.^{4,8,26,30} Chronically demyelinated and remyelinated nerve fibers form onion bulbs, identical to but less numerous than in hereditary demyelinating polyneuropathy (Charcot-Marie-Tooth disease type I). Routine toluidine blue, semi-thin plastic sections typically show an excess of large, thinly myelinated nerve fibers usually associated with varying degrees of acute axonal degeneration and reduced axonal density. Because CIDP is a multifocal process, nerve biopsy often does not yield a confirmatory diagnosis; recent large series have demonstrated inflammatory changes in 10% to 50% of specimens,^{1-5,26,83,84} axonal degeneration only in 20% to 40%, and normal biopsies in approximately 20% of cases.^{3,26,84}

The inflammatory infiltrates are composed primarily of CD8⁺ and CD4⁺ lymphocytes and macrophages within the endoneurium.^{8,86} Analysis of teased fibers is probably

the most sensitive method of demonstrating demyelinating changes, which are found in 50% to 80% of nerve biopsy specimens.^{4,8,83–87} However, the procedure is cumbersome and time-consuming and is performed primarily at specialized centers, which might be problematic for some clinicians as a result of regional availability. The use of nerve biopsy in CIDP continues to engender controversy; many experts endorse the value of a nerve biopsy, and especially teased fiber studies, in patients with suspected CIDP,^{8,83,84} whereas others have shown that the procedure adds little if other clinical and electrodiagnostic features support the diagnosis.^{30,87,88} However, one study, not surprisingly, has suggested that the presence of axonal loss on nerve biopsy implies a less favorable response to treatment.⁸⁹

With these ambiguities, it has been our practice to biopsy an electrophysiologically affected nerve (usually sural or radial sensory) in patients who are disabled by the neuropathy but do not fulfill electrodiagnostic criteria for CIDP, in those who have a concurrent illness that might confound the diagnosis (eg, patients with diabetes mellitus, inherited neuropathy) or influence treatment decisions (eg, patients with CIDP and HIV or hepatitis C infection), in those with an unusual variant (especially a sensory ataxic variant), and in patients who have failed to improve after treatment with the proven therapies (corticosteroids, IVIG, and plasma exchange) before embarking on more aggressive and potentially toxic treatment regimens. We apply the same principles to patients with paraprotein-associated neuropathies, and tend to obtain nerve biopsies only in those patients with unusual clinical features or who lack demyelinating changes on EMG and have a substantial functional deficit that might improve with therapy.

Differential Diagnosis

Several chronic demyelinating neuropathies might simulate CIDP and should be excluded before offering immune therapy. In many cases, the alternative demyelinating neuropathy also could be immune-mediated, and therefore some of the treatments might overlap with therapies for CIDP, but important distinctions exist (eg, the lack of response to corticosteroids in patients with multifocal motor neuropathy and anti-MAG neuropathy). In patients younger than 40 years, it would be prudent to exclude hereditary demyelinating neuropathy (CMT1A, CMT1B, CMTX) and hereditary liability to pressure palsy (HNPP). The latter disorders usually manifest in adolescence or young adulthood, and most patients have a positive family history and slower progression than CIDP. High arches, hammertoes, and palpably enlarged nerves are characteristic findings on examination, and, in contrast to CIDP, there is uniform conduction slowing in patients with CMT and generally no conduction block or temporal dispersion. DNA analysis confirms the presence of CMT1A, 1B, CMTX, and HNPP, and these studies should be per-

formed when hereditary demyelinating neuropathy is suspected.

Acquired demyelinating neuropathies that should be distinguished from CIDP include IgM-MGUS-associated neuropathies with anti-MAG or anti-GD1b antibodies as noted previously. Many experts have stressed that in IgM-MGUS neuropathy, the differences in the clinical presentation (distal, predominantly sensory pattern), electrophysiological features (distal demyelination; conduction block is less common), and lack of response to certain conventional therapies (eg, corticosteroids are generally ineffective) warrant a distinction from idiopathic CIDP.^{30,47,90} For example, most patients with IgM-MGUS neuropathies, especially with elevated anti-MAG antibodies, have substantially prolonged distal motor latencies.^{54–56} Similarly, multifocal motor neuropathy is clearly a separate entity, as a result of the focal or multifocal pattern, upper limb predominance, restricted involvement of motor fibers, and lack of response to corticosteroids in most cases.^{30,40,41} Toxin and medication-induced demyelinating neuropathies (due to solvents [“glue-sniffing neuropathy”], arsenic poisoning, amiodarone, and so on) also should be excluded.

Treatment

The primary goal of treatment is to reduce symptoms (weakness, sensory loss, imbalance, and pain), improve functional status, and, if possible, maintain long-term remission in patients with CIDP.¹ Conventional therapy for CIDP has included corticosteroids, plasma exchange (PE), and IVIG. Improvement can be expected in 60% to 80% of patients with one of these treatments.^{2,3,6,14–22,91,92} Each of these treatments has been demonstrated to be superior to placebo in randomized, double-blind, placebo-controlled studies, for which reason insurance reimbursement in the United States should no longer be a problem.^{14–19,21} Furthermore, randomized, prospective clinical trials have shown that plasma exchange and IVIG, and IVIG and prednisolone have similar short-term efficacy.^{20,22} The practical issues associated with these agents are the lack of a durable response in many patients and the difficulties of using these therapies, ideally suited for short-term administration, for long-term management of a chronic disease. IVIG is expensive, time-consuming, and might have limited availability; plasma exchange can be invasive for those who require central venous catheters and requires well-trained personnel at specialized centers, and corticosteroids are often poorly tolerated when used as a long-term therapy.⁹³

Prednisone

There are numerous anecdotal reports demonstrating the efficacy of prednisone in CIDP, with response rates ranging between 40% and 95%.^{3,15,26,94} In a 3-month, randomized, placebo-controlled trial of alternate-day,

high-dose prednisone (120 mg) in 28 patients, corticosteroids were demonstrated to be more effective than placebo.¹⁴ The effect was similar between patients with a progressive and relapsing course. The average time to induce a response with prednisone (60 mg/day) has been reported to be approximately 2 months with maximal improvement not observed until after 6 months.²⁶ Hughes and colleagues showed improvement in disability scores within 2 weeks of starting 60 mg of daily prednisolone.²² A minority of patients have remained in remission after tapering steroids, but most (up to 70%) have relapsed after discontinuing treatment and have required repeated courses of prednisone or alternative immunotherapy.^{3,94} In some patients, the addition of azathioprine or other immunosuppressive agents might sustain a remission and reduce the requirement for high-dose prednisone,^{1-2,5,26,27} but this benefit has not been confirmed in a randomized, controlled trial.⁹⁵ Young age at onset, duration of symptoms less than 6 months, slight neurologic impairment, and mild slowing of nerve conduction velocities have been associated with a favorable response to corticosteroids. A treatment period of less than 6 months and rapid tapering of prednisone might increase the risk of relapse.⁹⁴

Most patients will observe improvement in strength and sensory symptoms after several weeks of treatment.^{3,22} Maximal improvement occurs after 3 to 6 months of therapy, and only a minority of patients experience further recovery after 6 months.^{26,94} The standard dosage of prednisone is 1.0 to 1.5 mg/kg per day administered as a single morning dosage. Monthly pulse intravenous methylprednisolone or pulsed oral dexamethasone might be an effective alternative regimen in patients who cannot tolerate daily prednisone.⁹⁶ Patients who experience substantial clinical improvement after taking prednisone daily might reduce the dosage by switching to an alternate-day regimen after 2 or 3 months and tapering by 5 to 10 mg on the alternate day, every 2 to 4 weeks, as gauged by the patient's clinical state. If the patient deteriorates, the dosage should be increased to the minimum dose that induces improvement. Anecdotal experience suggests that adding an alternative treatment (PE, IVIG, azathioprine, cyclophosphamide, or cyclosporine A) should be considered in patients who develop adverse effects, relapse repeatedly following reduction of the prednisone dosage, or require more than 35 mg prednisone every other day.^{4,5,8,17,21,26,27,92-94,97-107}

It is our view that prednisone is best suited for younger patients with CIDP who have few other medical illnesses. Long-term prednisone therapy is generally poorly tolerated in the elderly and those who have concurrent medical conditions (eg, hypertension, diabetes, and obesity). Weight, blood pressure, blood glucose, electrolytes, ophthalmic evaluation, and stool guaiac require regular monitoring. In addition, periodic bone density measurements and a regimen to prevent steroid-induced osteoporosis should be instituted. This might include calcium and vitamin D, cy-

clical etidronate, calcitonin, or bisphosphonates. Contraindications to prednisone therapy include active peptic ulcer disease, poorly controlled diabetes mellitus or hypertension, severe osteoporosis, and systemic fungal infections.

The major limitation of prednisone therapy is the large number of well-known and potentially serious adverse effects that could develop. Acute side effects are often minor and include mood lability, insomnia, and fluid retention; steroid psychosis is a serious problem that could develop after acute treatment with high doses. Chronic complications of corticosteroids (usually occurring after several months of treatment) include: weight gain, hyperglycemia, gastritis, osteoporosis, aseptic necrosis of the femoral or humeral heads, vertebral fractures, steroid myopathy, impaired wound healing, increased susceptibility to infections (herpes zoster, cutaneous fungal infection, reactivation of tuberculosis, recurrent pneumonia), cataracts, glaucoma, hirsutism, cushingoid state, and pseudotumor cerebri.

Prednisone is inexpensive (approximately \$20/mo) and cost-effective in the short-term management of CIDP; the low cost and ease of administration are the best arguments in favor of the use of corticosteroids as initial therapy. However, many serious complications are associated with long-term corticosteroid use for which treatment might be expensive (eg, peptic ulceration with gastrointestinal bleeding, aseptic necrosis of the hip, sepsis, and so on), thus contributing to potentially large "hidden" costs associated with prolonged prednisone therapy.^{20,22} These hidden costs are unfortunately difficult to measure in any systematic fashion.

Plasma Exchange

Two randomized, placebo-controlled trials have demonstrated that PE is superior to placebo with response rates ranging between 33% and 80%.^{16,17} Improvement was observed in the mean Neurologic Disability Score, grip strength, and clinical disability grade; mean motor potential amplitudes and conduction velocities also increased after treatment. Patients improved within 4 weeks after initiating therapy, and those with a chronic progressive course apparently responded as well as patients with relapsing disease. Although the short-term efficacy of PE has been convincingly demonstrated, 50% to 67% of patients have deteriorated weeks to months after treatment and have required repeated exchanges or an alternative therapy to maintain improvement.^{3,17,92} In these patients, combining prednisone or other immunosuppressive agents with PE might induce a prolonged remission.^{7,8,97} Those with a short duration of disease and EMG or pathologic features of primary demyelinating neuropathy (without axonal loss) are more likely to respond to PE.^{17,89,98} Dyck and coworkers have established that the short-term efficacy of plasma exchange is similar to IVIG.²⁰

The beneficial effect of PE presumably results from the removal of pathogenic humoral factors such as circulating

immunoglobulins or autoantibodies. A single exchange removes 3 to 5 L of plasma and reduces intravascular IgG by approximately 45% so that 3 to 5 exchanges are required to remove approximately 90% of IgG. The exchanged plasma is replaced with 5% albumin and saline, sometimes supplemented with calcium, and with acid-citrate dextrose or heparin.

Although no specific guidelines have been established for the frequency or schedule of treatment, the frequency of PE can be guided by the duration of a clinical response. For example, those with severe disability (eg, nonambulatory) generally are treated with 5 exchanges (each exchange 250 mL/kg) performed over a period of 7 to 10 days. Improvement usually occurs within several weeks followed by a stable phase of weeks to months; exchanges can be repeated periodically in a similar fashion as necessary to maintain improvement. The addition of prednisone or other immunosuppressive agents (azathioprine or cyclosporine A) might sustain longer periods of remission and reduce the frequency of exchanges.^{8,17,92,103} Those who have mild or moderate disability and remain ambulatory might benefit from 2 or 3 exchanges per week over a period of 2 to 3 weeks, followed by once or twice weekly exchanges for an additional 3 weeks.¹⁷ PE can be used safely as a long-term therapy (years) in patients with a chronic relapsing course who do not respond or develop adverse effects with other treatment modalities.^{1,92} PE might be an especially appealing therapy for patients who have contraindications to other treatments such as prednisone for patients with diabetes mellitus and IVIG for those with preexisting renal insufficiency.

Patients with poor peripheral venous access might require placement of a central venous catheter with the associated risk of pneumothorax, catheter site infection, and venous thrombosis. The placement of a permanent central venous catheter facilitates frequent exchanges in those patients who have poor peripheral venous access. Intravenous saline (0.5–1.0 L) administered several hours before an exchange reduces the risk of hypotension and vasovagal symptoms. PE is generally well tolerated and safe but requires special equipment and well-trained personnel in medical facilities familiar with the procedure.

There are relatively few contraindications to PE, which include coagulopathy, thrombocytopenia, and hypotension. Complications of PE are: hypotension, cardiac arrhythmia, vasovagal reaction, citrate toxicity with nausea and vomiting, electrolyte imbalance (citrate-induced hypocalcemia), allergic reactions to infused plasma or plasma substitutes, anemia, thrombocytopenia, muscle cramps, paresthesias, and infection, bleeding, venous thrombosis, and pneumothorax as a complication of obtaining venous access.

Plasma exchange is an expensive procedure (approximately \$5000–10,000/series of exchanges) and is comparable to the cost of IVIG therapy in some regions. The expense might be offset by the benefit gained in work

productivity and the lack of serious long-term medical complications.

Intravenous Immune Globulin

IVIG has been introduced as the main therapy for CIDP over the last decade. Although early studies showed variable results,^{90,108} 4 well-controlled studies have demonstrated that approximately 60% to 70% of patients respond to IVIG.^{18–22} Improvement occurs within a few weeks, and rarely recovery might be dramatic, appearing 1 or 2 days after completing the infusion. Usually the benefit is transient (1–6 wk) with 50% of patients relapsing within weeks to months and subsequently requiring regular infusions to maintain maximum improvement.^{3,92,108} Patients with a progressive course or predominantly sensory deficits with tremor might be less likely to improve.²¹ Mendell and coworkers showed that 11 of 29 treatment-naïve patients with CIDP improved 1 or more disability grades after receiving IVIG, 1.0 g/kg per day at baseline and after 3 weeks, compared with 2 of 21 patients who received placebo; 76% of IVIG-treated patients had improved strength.¹⁸ Improvement was noted as early as 10 days after therapy. IVIG can be considered the initial treatment of choice for patients with CIDP because it has efficacy similar to the other first-line therapies, a lower frequency of adverse effects, and is easy to administer.^{18–22} Like with PE, anecdotal reports suggest that combining IVIG with prednisone or other immunosuppressive medications might increase the duration of remission and reduce the frequency of IVIG infusions.²¹

As previously noted, one study compared IVIG with prednisolone and demonstrated similar benefit after 6 weeks, with slight but not significant improvement in disability grade and grip strength favoring IVIG.²² A small, randomized crossover trial demonstrated that IVIG and PE induce a comparable degree of improvement in patients with CIDP and are probably equivalent in terms of efficacy and cost.²⁰ There have been no prospective, controlled trials comparing steroids, PE and IVIG; however, retrospective studies suggest that the rate of response is similar among these therapies.^{3,92} Uncontrolled studies have indicated that most patients who fail to improve with initial therapy will respond to an alternative treatment modality.^{3,8,20,92}

A single infusion of IVIG is derived from plasma from approximately 5000 human blood donors. The mechanism of the beneficial effects of IVIG is not precisely known but is thought to be related to modulation of pathogenic autoantibodies.¹⁰⁹ Other considerations include suppression of pathogenic cytokines, Fc receptor blockade, and alteration in T-cell function.¹⁰⁹

The standard dosage is 2.0 g/kg administered intravenously over 2 to 5 days. For patients who relapse and require periodic treatment at intervals from 1 to 4 weeks to maintain improvement, we usually offer a trial of oral immunosuppressants to decrease the frequency of IVIG ad-

ministration. Once the patient has stabilized, the dose of IVIG should be reduced and treatment interval increased as warranted by the patient's clinical status.

IVIG also can be administered by rapid infusion (1.0 g/kg per day over 4–6 h for 2 days) in younger patients and is generally well tolerated.¹¹⁰ The drug should be given at a slower rate (<200 mL/h) to the elderly or those with vascular disease, cardiac or renal insufficiency. Headache, fluid overload, and flu-like symptoms are more likely to occur at higher infusion rates. Pretreatment with 60 mg of intravenous methylprednisolone might reduce the severity of headaches in headache-prone patients. The first treatment should be monitored in a supervised setting (hospital, office, or outpatient clinic), and if well tolerated, subsequent infusions can be provided safely in the home. Absolute IgA deficiency is a risk factor for anaphylaxis after IVIG infusion, but it is so rare that routine screening is not necessary. However, renal function studies should be checked before IVIG administration, and the drug should be administered to patients with renal impairment only under close supervision. Elderly patients with borderline renal function are at especially high risk for renal failure. Although IVIG might cause worsening renal function in patients with diabetes mellitus, several studies have suggested IVIG is beneficial in patients with diabetes with CIDP.^{64,111,112}

IVIG is contraindicated in patients with absolute IgA deficiency or a history of a previous allergic reaction after exposure to human immune globulin. Minor side effects are common and include headache, malaise, fatigue, and fever.^{109,110,113,114} Other adverse effects include aseptic meningitis, rigors, myalgias, flushing, fluid overload with edema or congestive heart failure, renal insufficiency (presumably resulting from hyperosmolarity), transient neutropenia, back, chest, leg, or abdominal pain, and eczematous rash. Rarely, IVIG induces a hyperviscosity syndrome with increased risk for deep venous thrombosis, myocardial ischemia, or stroke. Revised manufacturing processes have largely eliminated the risk of viral transmission.

IVIG is expensive (\$8500–20,000/infusion) but the cost is comparable to PE.²⁰

Alternative Immunosuppressive Regimens

It is our opinion that the indications for alternative immunosuppressive treatment of patients with CIDP are: 1) the patient has not improved with sequential or combined trials of the conventional therapies we have outlined here that have demonstrated efficacy in randomized, controlled trials; 2) the patient has improved with these treatments but has frequent relapses, making continued therapy cumbersome, time-consuming, and costly; or 3) the patient has developed intolerable adverse effects. The therapies listed subsequently are promising and have been reported to be effective in some patients with CIDP in single case reports,

case series, and retrospective reviews, but have not been submitted to rigorous clinical trials that establish efficacy. Accordingly, it is prudent to advise the patient that these agents are not likely to be more effective than proven treatments and to maintain realistic expectations about recovery; this is particularly true in patients with concomitant severe axon loss in whom the neurologic deficit might be irreversible. **In those individuals, a reasonable expectation would be to stabilize the condition and halt further progression.**

Azathioprine

Although anecdotal reports have indicated that azathioprine is an effective therapy for CIDP,^{5,26,27,99,100} one unblinded but randomized study of only 9 months duration showed that the degree of improvement with azathioprine combined with prednisone was similar to prednisone alone.⁹⁵ However, the immunosuppressive effects of this agent might not occur for 6 to 12 months,^{5,27,99–101} and therefore the duration of this trial was too short to exclude benefit. Others have found that azathioprine is helpful in some patients not only for stabilizing the disease course, but more importantly, for allowing reduction of prednisone dosage.^{8,101} Therefore, azathioprine is used ideally as adjunctive therapy for CIDP as a steroid-sparing agent. The usual dosage is 2.0 to 3.0 mg/kg per day administered orally as a single daily dose. A test dose of 50 mg per day can be given for the first week. If this is tolerated, the dose is gradually increased by 50 mg every few days. It has been recommended that a complete blood count and liver enzymes be monitored every week for 4 weeks, then monthly for 6 months, and then every 3 months; more frequent monitoring is required in patients with leukopenia, thrombocytopenia, or liver enzyme abnormalities. Dose adjustments might be required based on these monitoring studies. Macrocytosis occurs commonly but does not require discontinuation of therapy. The blood count and liver enzymes should be monitored as long as the patient remains on azathioprine because toxic effects on the bone marrow and liver might be delayed.

Contraindications to azathioprine include severe leukopenia (WBC count <1.0 K/uL), thrombocytopenia (platelet count <50 K/uL), and systemic infection. Allopurinol inhibits the principal pathway for detoxification of azathioprine; therefore, the dosage of azathioprine should be reduced to one third to one fourth of the usual dosage when administered concurrently with allopurinol. Angiotensin-converting enzyme inhibitors should be avoided in patients treated with azathioprine as a result of an increased risk of severe leukopenia.

An acute hypersensitivity reaction might occur in the first several days to weeks of therapy characterized by severe nausea and vomiting, diarrhea, fever, malaise and myalgias, rash, elevation in liver enzymes, and pancreatitis; these adverse effects are reversible when the drug is discontinued.

Delayed side effects include dose-dependent bone marrow

suppression with leukopenia and thrombocytopenia, reduced resistance to infection, neoplasia, hepatotoxicity manifested by elevated liver enzymes, rash, alopecia, fever, and arthralgias. The risk of neoplasia is another concern in patients on long-term azathioprine therapy.¹¹⁵ Azathioprine is potentially teratogenic, and women should be advised to use contraceptive measures during and at least 6 months after discontinuing treatment.

Cyclosporine A

Patients with refractory CIDP might improve with cyclosporine A. Although published experience with this drug is restricted to several small case series, response rates have ranged from 40% to 90%. Patients improved within weeks of initiating treatment, and several discontinued prednisone without clinical deterioration.^{105–107} The initial dosage is 5.0 mg/kg per day orally in 2 divided doses 12 hours apart; dose adjustments are made by following target blood levels. Once the patient has stabilized, the dosage should be titrated to the lowest dose required to maintain improvement. A clinical response might be seen within several weeks after initiating therapy.¹⁰⁷ Cyclosporine A is contraindicated in patients with systemic infection, a history of previous hypersensitivity reaction, abnormal renal function, uncontrolled hypertension, and malignancy. Blood pressure, renal and liver function (including creatinine clearance), serum lipids, magnesium, and potassium require regular monitoring. Checking trough serum cyclosporine levels could minimize toxicity. The recommendation is for a trough level between 100 and 200 ng/mL. The following are indications for dosage adjustment or discontinuation: a rise in serum creatinine level of greater than 50% above the baseline or an absolute creatinine of more than 2.0 mg/dL, a rise in systolic blood pressure of greater than 150 mm Hg or diastolic blood pressure greater than 90 mm Hg, and a trough serum level higher or lower than recommended values based on the clinical response. Dose adjustments should not exceed more than 1.0 mg/kg per day and not occur more frequently than every 4 weeks.

The main side effects of cyclosporine A are renal insufficiency, hypertension, hirsutism, tremor, gingival hyperplasia, increased risk of opportunistic infection, sepsis, nausea, vomiting, hepatotoxicity, seizures (possibly related to hypomagnesemia), headache, and cramps.

The major difficulties with routine use of cyclosporine A are related to the large number of drug interactions that occur when used with other medications. For example, cyclosporine A should not be used with digoxin (increased risk of severe digoxin toxicity) or potassium-sparing diuretics (risk of hyperkalemia).

Cyclophosphamide

Oral cyclophosphamide also has been reported to be beneficial for patients with CIDP when administered for sev-

eral months as monotherapy or in combination with prednisone.^{5,89,102,103,116} Uncontrolled case series suggest monthly, pulse intravenous cyclophosphamide is effective when combined with prednisone or administered after cycles of PE.^{1,99,103,104} Good and coworkers reported that 12 of 15 patients with CIDP who were refractory to plasma exchange, IVIG, or steroids improved with monthly intravenous pulse cyclophosphamide (1 g/m²); the average time to sustained improvement was 8.5 months.¹¹⁶ Complications included nausea, headache, rash, and an anticipated leukopenia. No patient developed severe bone marrow suppression or malignancy with an average follow up of 3 years. Other reports have highlighted a treatment response following high-dose cyclophosphamide and myeloablative chemotherapy with autologous stem cell rescue.^{117,118} Cyclophosphamide is reserved for patients with CIDP who fail conventional immunotherapy and is most often administered in conjunction with prednisone, IVIG, or PE.^{5,27,97,101–104,116,117} The drug can be given orally as a single daily dose or as pulse intravenous therapy on a monthly basis. The response rate has been reported to be as high as 80%; however, experience is restricted to retrospective case series.^{89,103,104,116,117}

Leukopenia is a common adverse effect and a complete blood count with differential, platelet count, and urinalysis should be checked 2 weeks after intravenous administration and every 2 to 4 weeks in patients taking oral cyclophosphamide. With intravenous therapy, the dosage should be adjusted by 25% increments to produce a transient reduction of the white blood cell count to 1.0 to 2.0 K/uL. Significant nausea can be avoided by administering ondansetron hydrochloride (8 mg orally twice a day) or other antiemetics, and at least 3 L of intravenous fluid should be given before and during the infusion to reduce the risk of hemorrhagic cystitis.

The standard oral dosage is 2.0 mg/kg per day; as mentioned, when administered intravenously, the typical dose is 1.0 g/m² administered monthly. The main side effects are dose-dependent bone marrow suppression, consequent reduced resistance to infection, hemorrhagic cystitis, infertility, teratogenic effects, amenorrhea, nausea, vomiting, anorexia, alopecia, and malaise. Clinicians should be aware of the increased risk of malignancies, especially lymphoma and leukemia, in patients with long-term (usually several years) exposure or high-dose cyclophosphamide therapy, and a history of previous bone marrow suppression or myelodysplastic syndrome is a contraindication to treatment with cyclophosphamide. It has been our experience that some patients have a modest leukopenia that persists for several months after discontinuing cyclophosphamide; for this reason, we often recommend drug “holidays” of 3 to 6 months following treatment with 6 months of pulse intravenous cyclophosphamide or 12 months of oral therapy.

Interferons

Interferons have complex immunomodulating effects and can influence the levels of proinflammatory cytokines, especially tumor necrosis factor alpha, interleukin-2, and interferon gamma, which have a role in the development of inflammatory demyelination. Several case reports and our prospective, unblinded pilot study indicated that interferon alpha 2a was effective in patients with CIDP who were refractory to conventional immunomodulating agents.^{119–122}

No randomized, controlled trials with this agent have been conducted. Similarly, initial reports indicated that interferon beta-1a was effective in several patients with CIDP who did not improve after trials of prednisone, plasma exchange, cyclosporine A, IVIG, and azathioprine.^{123,124} A randomized, double-blind crossover trial of 10 patients with CIDP showed that interferon beta-1a was not superior to placebo.¹²⁵ However, that study was not powered to assess efficacy and the duration of therapy was limited to 12 weeks.¹²⁵ In contrast, a prospective, multicenter, open-label trial of interferon beta-1a (30 µg intramuscularly weekly) for 6 months in 20 patients with CIDP showed significant improvement in the neurologic disability score and clinical grading scale compared with baseline measures.¹²⁶ Thirty-five percent of patients improved one or more functional grades, and the drug was safe and well tolerated.¹²⁶ Particularly because these patients were refractory to or dependent on other agents, the authors have suggested that these promising results warrant a larger placebo-controlled trial (currently planned).

The usual dosage of interferon alpha 2a is 3 million IU subcutaneously 3 times per week, and for interferon-beta 1a, 3 to 10 million IU subcutaneously 3 times per week or 30 µg intramuscularly weekly, depending on the formulation. Common side effects include influenza-like symptoms (fever, chills, sweating, myalgias), headache, fatigue, dizziness, nausea and vomiting, diarrhea, anorexia, abdominal pain, depression, irritability, and injection site reaction.

Mycophenolate Mofetil

Mycophenolate Mofetil (MM) is a novel immunosuppressant that been used successfully for treatment of acute rejection after renal transplantation and might be appropriate for the treatment of immune-mediated neuromuscular diseases.^{23,24,127} MM is an inhibitor of the *de novo* pathway of purine nucleotide synthesis, thereby blocking the proliferation of B- and T-cell lymphocytes.¹²⁷ MM produces immunosuppression that is comparable to azathioprine without causing major bone marrow suppression. Chaudhry and associates reported that 1 of 3 treated patients with CIDP had improved strength and was able to reduce the methylprednisolone dosage.²³ Another group reported benefit in 2 patients who were refractory to other treatments.²⁴ The onset of improvement occurred within 3 to 4 months. We have used MM in 10 patients with CIDP

over a 3-year period.¹²⁸ Three patients (30%) improved. One patient had moderate disability as a result of generalized weakness (Rankin 3) and recovered normal strength and improved sensory function after 4 weeks of treatment. Another was able to lengthen the interval of IVIG treatment from every 6 to 9 weeks. The third patient with CIDP had severe, generalized weakness (Rankin 4) and recovered fully after 4 months of prednisone and MM; prednisone was discontinued and the patient remained stable on MM for 16 additional months. We have concluded that MM can provide modest benefit to a minority of patients with CIDP, usually by stabilizing their condition and allowing reduction of steroid or IVIG therapy.¹²⁸ Larger randomized, controlled trials might further clarify the role of this agent in CIDP.

MM has been administered in combination with cyclosporine A and prednisone. The efficacy and safety of mycophenolate mofetil in combination with other immunosuppressants has not been determined. The standard dosage is 1.0 g administered twice per day, similar to the dosage that is recommended for renal transplant patients. Patients should be monitored for neutropenia with complete blood counts monthly for 6 months and then every 3 months thereafter.

This agent should not be used in pregnant women because of its teratogenic effect in animals. Similarly, myelodysplasia with persistent neutropenia is a relative contraindication to the use of MM. Potential side effects include nausea, headache, tremor, susceptibility to infection, leukopenia, and possibly an increased risk of developing lymphoma and other malignancies.¹²⁷

Total Lymphoid Irradiation

This standard treatment of Hodgkin disease has been shown to induce profound, long-lasting suppression of cell-mediated immunity and could be effective in CIDP, but efficacy data are limited to one small uncontrolled case series.¹²⁹ No specific guidelines for the administration of total lymphoid irradiation (TLI) for CIDP have been established. One study used irradiation in a “mantle” field of 200 rads per day 5 days per week for a dose of 2000 rads followed by further irradiation in an “inverted Y” field using the same protocol for an additional 2000 rad.¹²⁹ We have no experience with TLI for CIDP, and this therapy probably should be reserved for severely disabled patients who are refractory to proven first-line therapies and combined therapy with cytotoxic agents.

Treatment Recommendations

What is the preferred first-line therapy? The “best” therapy for CIDP has not been established. The consensus has been that corticosteroids, IVIG, and plasma exchange are probably equivalent. IVIG is easy to administer and generally well tolerated, and therefore is the preferred initial therapy by many experts. However, many patients relapse

and require repeated IVIG infusions, leading to higher costs and a burden for the patient who is committed to long-term IVIG therapy. Prednisone might be well tolerated in younger individuals and those without obvious contraindications (eg, diabetes, osteoporosis, peptic ulcer disease, and so on), and is inexpensive and easy to administer. However, long-term corticosteroid therapy brings complications, and the majority of patients with CIDP eventually require concomitant therapy with a steroid-sparing agent (azathioprine, cyclosporine, mycophenolate mofetil, and so on) to allow for a tapering of corticosteroids without a clinical relapse. Patients who are severely disabled (wheelchair- or bed-bound) at presentation and do not have prominent axonal loss on EMG would be good candidates for plasma exchange or IVIG because of the desire for a rapid clinical response. Prednisone can be added later to enhance further recovery during the ensuing months.

Approximately one third of patients with CIDP do not respond to conventional therapies. In these patients, it is important to confirm the diagnosis if any uncertainty remains, and a nerve biopsy should be considered. Switching to an alternative first-line therapy might yield a higher response rate. In patients who fail 2 conventional therapies, the third should be offered. Thereafter, second-line therapies (azathioprine, cyclosporine A, cyclophosphamide, interferon, mycophenolate mofetil) can be administered with the selection of a treatment regimen individualized to consider potential side effects, patient convenience, and cost.

What is the treatment of patients with CIDP and electrophysiological signs of substantial axonal loss? Axonal regeneration is a much slower process than remyelination. Accordingly, clinical improvement in these patients is frequently prolonged and usually incomplete. For example, patients with generalized proximal and distal weakness might have meaningful improvement in proximal strength after standard treatments (eg, regain the ability to walk), but severe hand weakness and foot drop remain. Our experience suggests that most patients with CIDP and substantial axonal loss require prolonged or repeated treatments for 6 to 12 months or more before assessing the efficacy of any intervention. Some experts endorse more aggressive cytotoxic regimens (eg, pulse cyclophosphamide), often combined with other regimens (prednisone, plasma exchange), but there is little or no published data to guide the clinician in this situation. At our center, we usually combine prednisone with plasma exchange (arbitrarily 5 exchanges over 2 weeks, then 2–3 exchanges monthly) or IVIG monthly for 6 months. If there is no improvement, we usually add cyclosporine or pulse intravenous monthly cyclophosphamide to the regimen. **As previously noted, in many individuals, the neurologic deficit might be irreversible and the goal of therapy should be to delay further progression.**

Supportive Therapies

Patients certainly benefit from walking sticks, walkers, ankle-foot orthotics, and other rehabilitation devices and strategies to assist ambulation and other activities of daily living. Physical and occupational therapy might be helpful to maintain range of motion, prevent joint contractures in paretic limbs, and assist in gait retraining. A moderate exercise regimen might help reduce physical fatigue and increase endurance.

Prognosis

Initial descriptions of this disease highlighted the substantial disability that accrued in most patients.⁴ With earlier recognition of the disorder by neurologists and more effective and aggressive management strategies, the overall prognosis is probably more favorable; with treatment, more than two thirds of patients remain independent and able to work.^{1–6,8,101,129,131,132} Others have reported that as many as 90% improve with therapy, but 50% relapse within a few years and fewer than one third remain in remission without therapy.^{3,26,130} In our experience, approximately 10% to 15% of patients have complete remission with a normal neurologic examination and no functional disability. Conversely, several large series have suggested that approximately 5% to 10% of patients die of their illness or the complications of treatment.^{3–5,26} Greater disability and a poorer long-term prognosis have been correlated with an older age of onset (>30 y), 4-limb weakness at onset, progressive course, central nervous system involvement, and prominent axonal loss on nerve biopsy.^{130,131}

Conclusion

Much more is known about treatment of CIDP than the earliest descriptions offered. However, the essential nature of the disease, the limits of the disorder, and the relation of CIDP to paraproteinemia and systemic diseases are based largely on clinical opinion. Further study of the immunopathogenesis of the illness is needed. In the meantime, reasonably good evidence guides current therapy, and the clinician must take a sequential heuristic approach to each patient.

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